

Shu Xie

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ResearchGate: <https://www.researchgate.net/profile/Shu-Xie-5>



EDUCATION

University of Groningen, Netherlands

2018.10 – 2024.3

Ph.D. in Theoretical Ecology

- Supervisors: Prof. Rampal Etienne and Dr. Luis Valente
- Thesis: Trait-dependent diversification in island biogeography.
 - Proposed a stochastic model to simulate community assembly and trait dynamics over time in insular systems.
 - Developed a computational pipeline to investigate the robustness of island biogeography models ignoring trait effects.
 - Developed Bayesian inference methods (MCMC and ABC) to estimate rates in diversification models.

China Agriculture University, China

2016.9 – 2018.6

M.S. in Grassland Restoration Ecology

- Supervisor: Prof. Kun Wang
- Thesis: Effects of interrow grass cultivation on soil physicochemical properties and grape quality in vineyards.
- Other project: Response of soil microbial biomass to climate and land use change (Global *meta*-analysis).

China Agriculture University, China

2012.9 - 2016.6

B.S. in Grassland Restoration Ecology

- Supervisor: Prof. Kun Wang
- Thesis: Allelopathic effects of *Orychophragmus violaceus* on the germination and growth of forage.

PUBLICATIONS

Peer-Reviewed Articles:

Xie, S., Valente, L., & Etienne, R. S. (2023). Can we ignore trait-dependent colonization and diversification in island biogeography? *Evolution*, 77(3), 670–681. <https://doi.org/10.1093/EVOLUT/QPAD006>

Li, S.*, Xie, S.*, Zhang, S., Miao, S., Tang, S., Chen, H. & Zhan, Q. (2022), Global patterns and controls of the soil microbial biomass response to elevated CO₂. *Geoderma*. 428 (8)., 116153. <https://doi.org/10.1016/j.geoderma.2022.116153>
(* equally contributed to this work).

Li, J., Xie, S., Wilson, G. W. T., Cobb, A. B., Tang, S., Guo, L., Wang, K. & Deng, B. (2019), Plant-microbial interactions facilitate grassland species coexistence at the community level. *Oikos*. 129 (4), 533-543.
<https://doi.org/10.1111/oik.06609>

Zhang, Y., Tang, S., Xie, S., Liu, K., Li, J., Chen Q., Huang D., Wang, K. (2019), Effects of species-dominated patches on soil organic carbon and total nitrogen storage in a degraded grassland in China. *PeerJ* 7:e6897
<https://doi.org/10.7717/peerj.6897>

Tang, S., Guo, J., Li, S., Li, J., Xie, S., Zhai, X., Wang, C., Zhang, Y., Wang, K. (2019), Synthesis of soil carbon losses in response to conversion of grassland to agriculture land. *Soil and Tillage Research* 185, 29-35.
<https://doi.org/10.1016/j.still.2018.08.011>

Preprints:

Xie, S., Valente, L., & Etienne, R. S. (2023). Identifying summary statistics for approximate Bayesian computation in a phylogenetic island biogeography model. (BioRxiv). <https://doi.org/10.1101/2023.10.13.562305>.

Xie, S., Valente, L., & Etienne, R. S. (2023). Inferring state-dependent diversification rates using approximate Bayesian computation (ABC). (BioRxiv). <https://doi.org/10.1101/2023.10.14.562317>.

PRESENTATIONS

Seminar talk: Can we ignore trait effects in island biogeography? Groningen, the Netherlands. (2023)

Conference talk: Approximate Bayesian Computation in island biogeography models. Mathematical Models in Ecological and Evolution (MMEE), Reading, the UK. (2022)

Poster: Effect of trait dynamics on diversification rates in insular systems. Early Career Biogeographers Conference (ECBC), Amsterdam, the Netherlands. (2021)

Poster: The robustness of a simple island biogeography model with trait effects. Meeting of the Dutch Society for Evolutionary Biology (NLSEB), Ede, the Netherlands. (2021)

TECHNICAL SKILLS

Methods: Maximum likelihood estimation; Theoretical modelling in ecology and evolution;
Bayesian inference using Markov Chain Monte Carlo and Approximate Bayesian Computation;
Phylogenetic comparative methods; Meta-analysis

Tools: R; C++; Mathematica; High-performance computer (HPC) cluster; SPSS; SAS

Field Work: Soil and plant biomass sampling

Lab Work: Spectroscopy; Gas chromatograph; Plant and soil physiochemical analysis

TEACHING EXPERIENCE

Teaching Assistant at the University of Groningen:

- Evolutionary Processes (Computer practical on trait evolution and island biogeography). (2020, 2021)
- Mathematics in the Life Sciences. (2019, 2020)
- Biological Modelling and Model Analysis (Computer practical). (2019, 2020)

HONORS AND AWARDS

- First-class outstanding student scholarship (2015, 2016, 2017)
- Excellent graduate of China Agriculture University (2016)
- CAU Merit Student (2015)
- The Jin Zhengda Scholarship (2015)
- National Encouragement Scholarship (2013, 2014)

SOCIETY MEMBERSHIP

Society for the Study of Evolution (SSE). (2021, 2023)

Dutch Society for Evolutionary Biology (NLSEB). (2021)

REFERENCES

Prof. Rampal Etienne

University of Groningen, Groningen, Netherlands

Email: r.s.etienne@rug.nl

Dr. Luis Valente

University of Groningen, Groningen, Netherlands & Naturalis Biodiversity Center, Leiden, Netherlands

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Prof. Kun Wang

China Agriculture University, Beijing, China

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